

BANARAS HINDU UNIVERSITY

B.A./B.Sc. (Hons.) Semester VI Examination, 2013-14

**Paper No. STA/STB-602: Demand Analysis, Analysis of Income
Distribution and Queuing Theory**

Time: 3 Hours

Maximum Marks: 70

*Instructions: Answer any **five** questions. All questions carry equal marks. Symbols have their usual meanings.*

Question 1.

- (a) Explain the concepts of supply, demand, and equilibrium point. Explain the meaning of a change in demand and a shift in demand with examples, and state the law of demand. [7]
- (b) Define elasticity. Explain price elasticity and income elasticity of demand. Classify goods on the basis of income elasticity of demand with suitable examples. Derive the demand function for the case of constant price elasticity. [7]

Question 2.

- (a) Explain Engel's Law and Engel's Curve, and critically comment on both. Explain the methods for empirically determining the Engel's curve. [7]
- (b) Explain Leontief's input-output model along with its assumptions, steps, and limitations. [7]

Question 3.

- (a) Explain Pareto's law of income distribution and the effect of income elasticity on Pareto's curve. Using Pareto's Law, find the frequency distribution $f(x)$, $a < x < \beta$, of income, where a is the lowest income and β is the highest income. [7]
- (b) Write the functional form of the Lorenz curve and describe the relation between the Lorenz curve and the Gini coefficient. Define the Gini coefficient along with its advantages and disadvantages as a measure of income inequality. [7]

Question 4.

Explain queuing theory with examples. Write down the different types of queuing disciplines used in queuing theory, with the help of real-life examples. [14]

Question 5.

- (a) Write down the assumptions for the number of arrivals in a queue. Show that the distribution of the number of arrivals in a queue follows the Poisson distribution. [7]
 - (b) Prove that in a Poisson arrival process, the inter-arrival time distribution follows the exponential distribution. [7]
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Question 6.

Explain the queuing model $(M/M/1) : (\infty/\text{FCFS})$ and obtain the steady-state condition (performance measures: L, L_q, W, W_q) for this model. [14]

Question 7.

- (a) Obtain the waiting time distribution for the queuing model $(M/M/c) : (\infty/\text{FCFS})$. [7]
 - (b) On average, 96 patients per 24-hour day require the service of an emergency clinic. On average, a patient requires 10 minutes of active attention. Assume the clinic can handle only one emergency at a time. Suppose it costs the clinic Rs. 100 per patient treated to obtain an average servicing time of 10 minutes, and each minute of decrease in this average time costs Rs. 10 per patient treated. How much should be budgeted by the clinic to decrease the average queue size from $\frac{11}{3}$ patients to $\frac{1}{2}$ patient? [7]
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Question 8. Explain any **three** of the following: [14]

- (a) Cross elasticity of demand
 - (b) Log-normal distribution
 - (c) Erlang distribution
 - (d) Transient and steady states in queuing theory
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